

DS 358 DSP - Nature of Speech Signal

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January 1, 2025



Some Resources for Deep Speech Processing

- L.R. Rabiner and R.W. Schafer, Digital Processing of Speech Signals Pearson Education, Delhi, India, 2004
- L. R. Rabiner, B. H. Juang and B. Yegnanarayana, "Fundamentals of speech recognition", Pearson Education, 2009.
- Linus Ericsson, et al., Self-Supervised Representation Learning: Introduction, Advances and Challenges, arXiv:2110.09327
- R. Prabhavalkar, et.al., "End-to-End Speech Recognition: A Survey," in IEEE/ACM Transactions on Audio, Speech, and Language Processing, vol. 32, pp. 325-351, 2024,
- Xie, Tianxin, et al. "Towards Controllable Speech Synthesis in the Era of Large Language Models: A Survey." (2024).
- Wang, S., et.al., "Overview of speaker modeling and its applications: From the lens of deep speaker representation learning" IEEE/ACM Transactions on Audio, Speech, and Language Processing.



- **Overview of Speech Processing:** Speech production, perception and comprehension, spoken language processing, human speech processing tasks, nature of speech signal, signal processing of speech, linguistics and phonetics, feature extraction. (6 hrs)
- **Machine Learning of Speech Processing:** clustering of speech features, classification tasks in speech, Gaussian mixture model (GMM), Hidden Markov Model (HMM), Support Vector Machine (SVM) and Neural Networks (NN) (6 hrs)
- **Deep Learning of Speech Processing:** Self-Supervised Learning for Speech Representation. Automatic Speech Recognition (ASR). Text to Speech (TTS) synthesis. Speaker and Language Recognition. (Details in Annexure - DSP) (30 hrs)

Summary from Introduction & Sound Units

- Human speech processing
- Speech as cognitive processing
- Speech as physiological processing
- Speech production, perception and comprehension
- Natural Language : Spoken vs written
- Information sources in speech
- Sounds units
- Classification of sound units

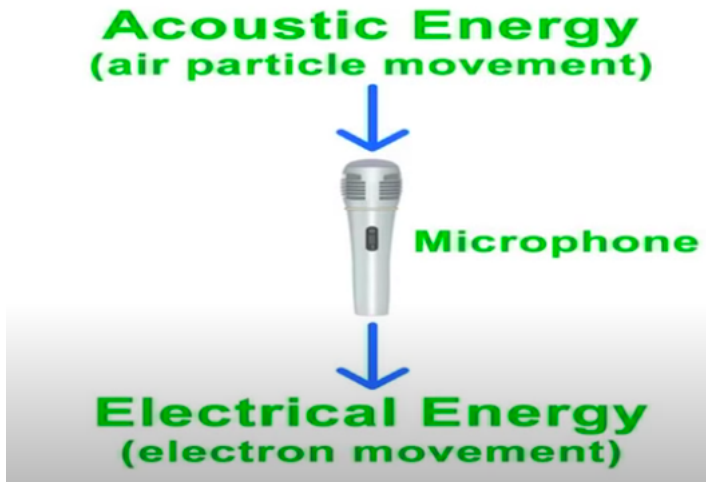
Organization

- Digital Speech - sampling freq and bit resolution
- Time and frequency dom representation of speech
- Waveform and spectrum of speech
- Voiced, unvoiced and silence regions of speech

Digital Speech - Sampling Frequency, Bit Resolution

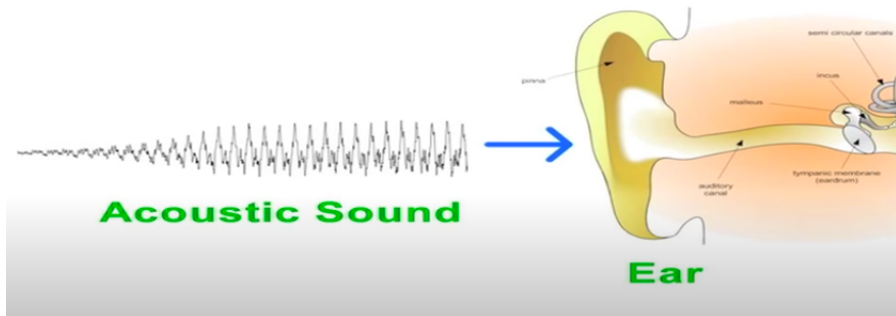
- Acoustic to analog electrical - transducer - microphone
- Electrical to acoustic - transducer - speaker
- Analog - discrete time signal
- Sampling process - sampling frequency
- Discrete - digital speech
- Analog to digital conversion - bit resolution

Acoustic to Electrical Conversion



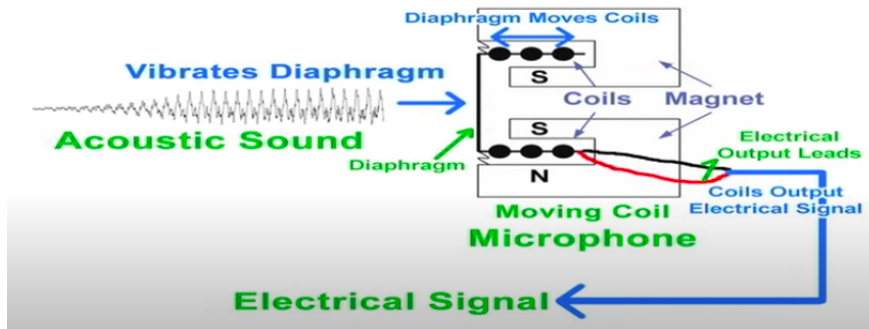
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Acoustic to Bio-Electrical Conversion



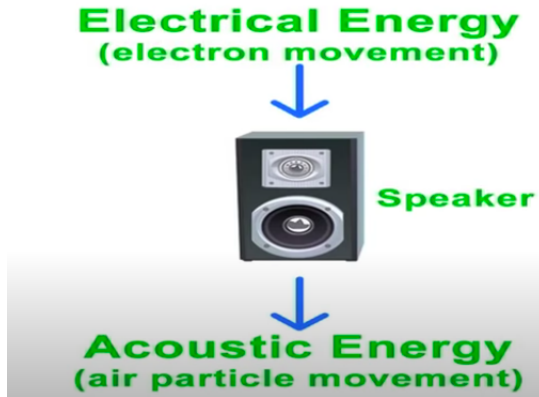
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Acoustic to Electrical Conversion



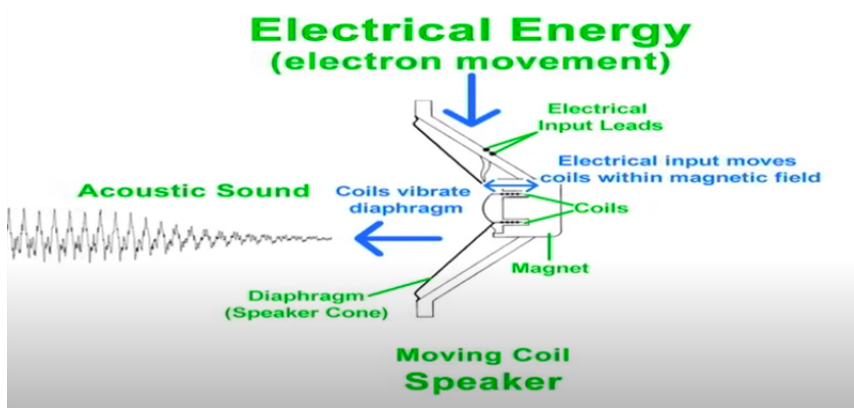
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Electrical to Acoustic Conversion



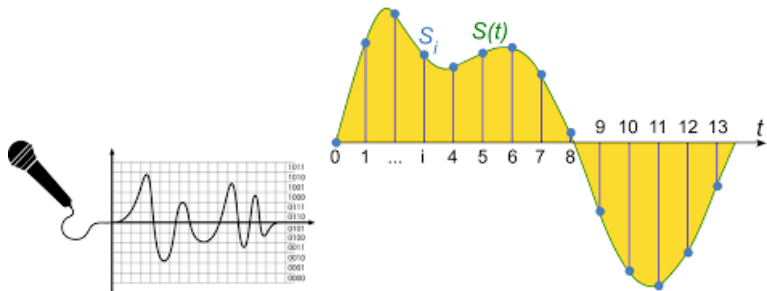
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Electrical to Acoustic Conversion



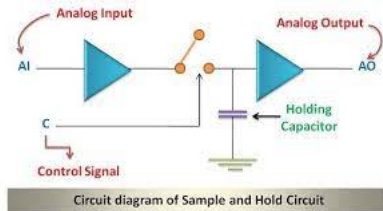
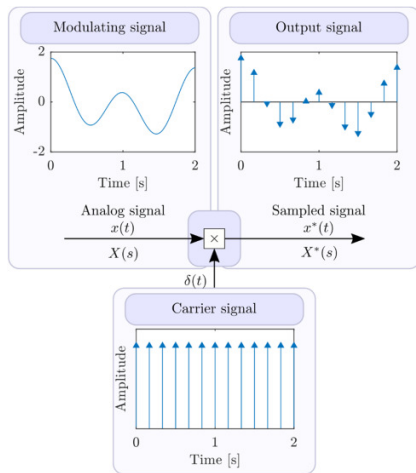
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Analog to Discrete



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Sampling Process

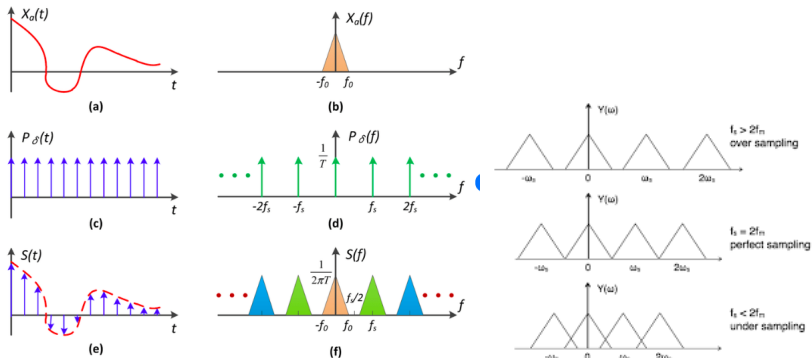


Electronics Coach

- Sampling Interval T_0

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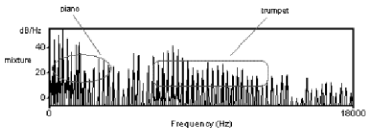
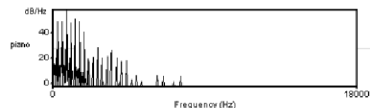
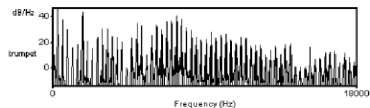
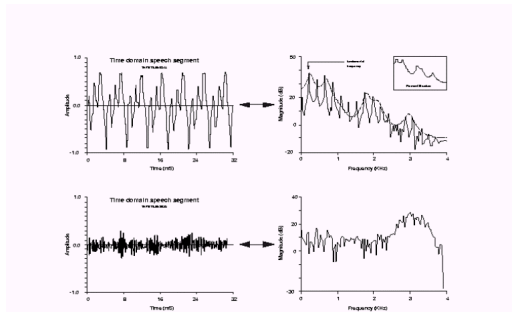
Sampling Theorem



- Sampling frequency $f_s \geq 2f_0$ or $\omega_s \geq 2\omega_0$
- Sampling period $T_0 = \frac{1}{f_s}$

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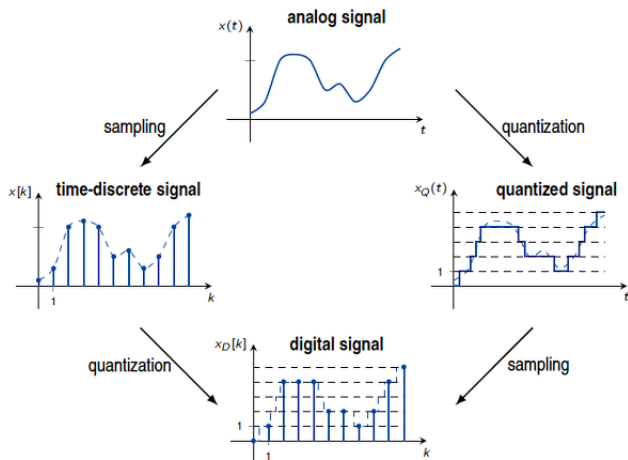
Significance of Sampling Frequency in Speech Processing



3. Power spectrum of trumpet, piano and their mixture; frequency axis

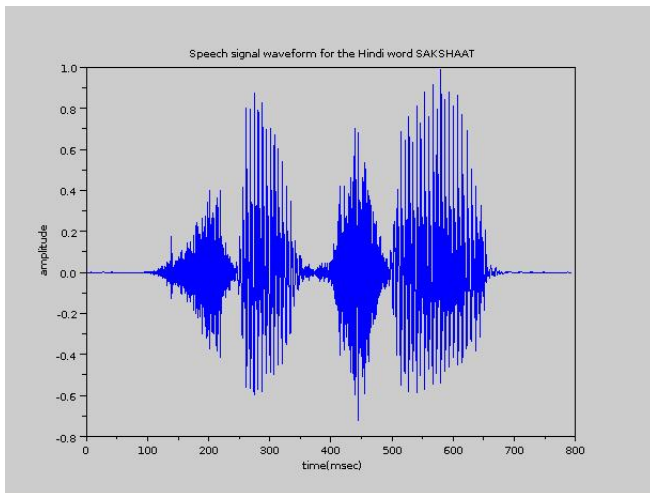
- 8 kHz for vowel-like sounds, 16 kHz for fricative-like sounds and 44 kHz for music

Discrete to Digital Conversion



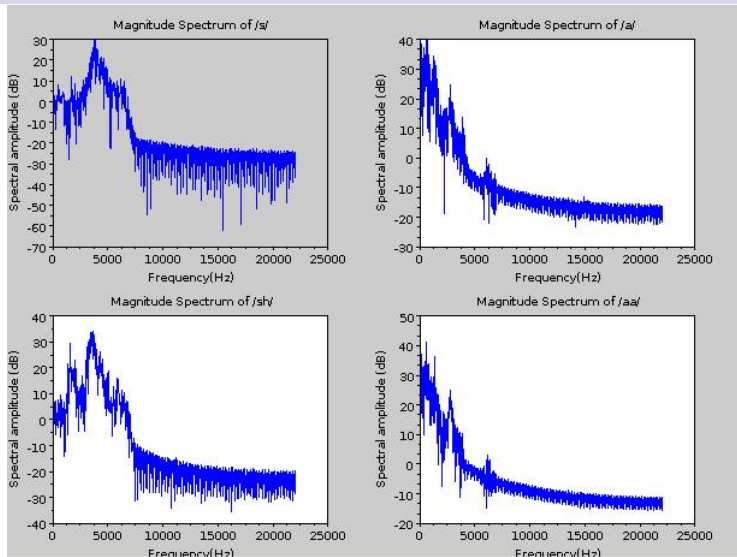
- No. of Quantization levels $q = 2^n$, n no. of bits in binary word (bit resolution)
- $n = 8 \implies q = 2^8 = 256$

Speech Signal Sampled at 44.1 kHz



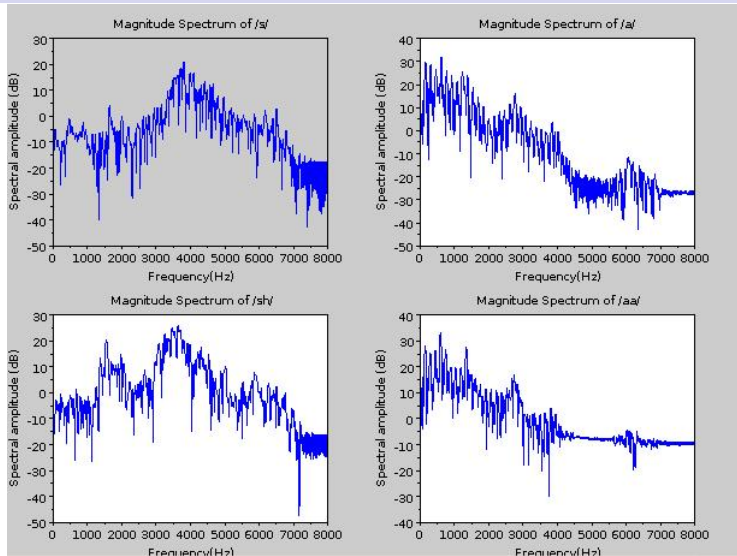
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Different Speech Segments Spectra Sampled at 44.1 kHz (1)



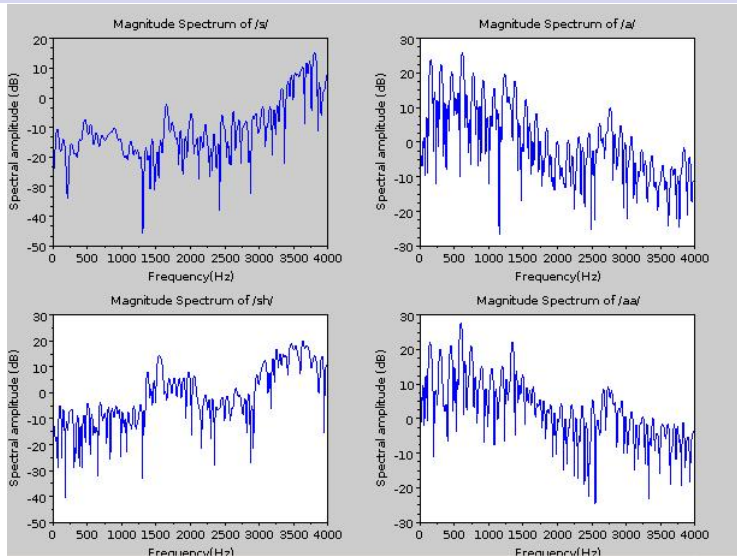
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Different Speech Segments Spectra Sampled at 44.1 kHz(2)



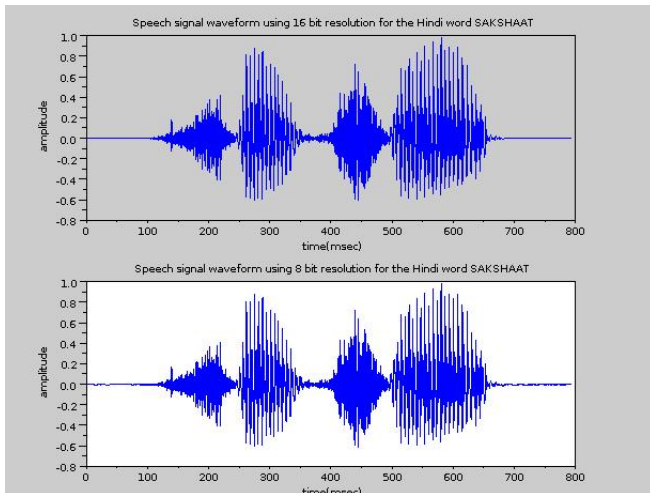
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Different Speech Segments Spectra Sampled at 8 kHz (Telephone / Mobile)



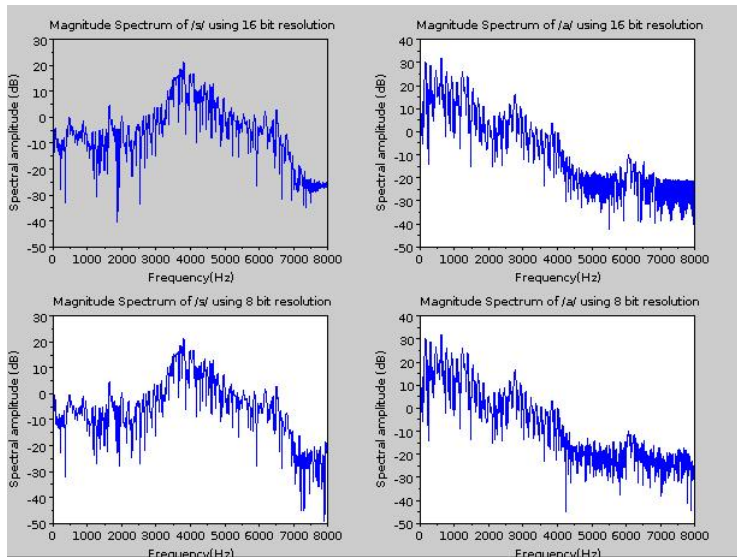
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Speech Signal at 16 vs 8 bit resolution



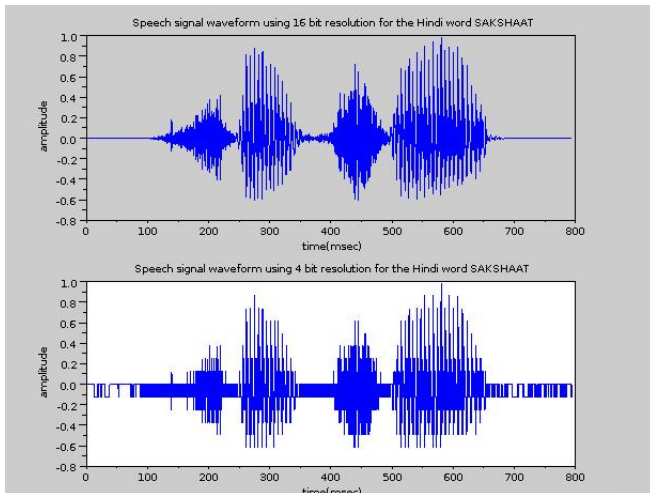
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Speech Spectra at 16 vs 8 bit resolution



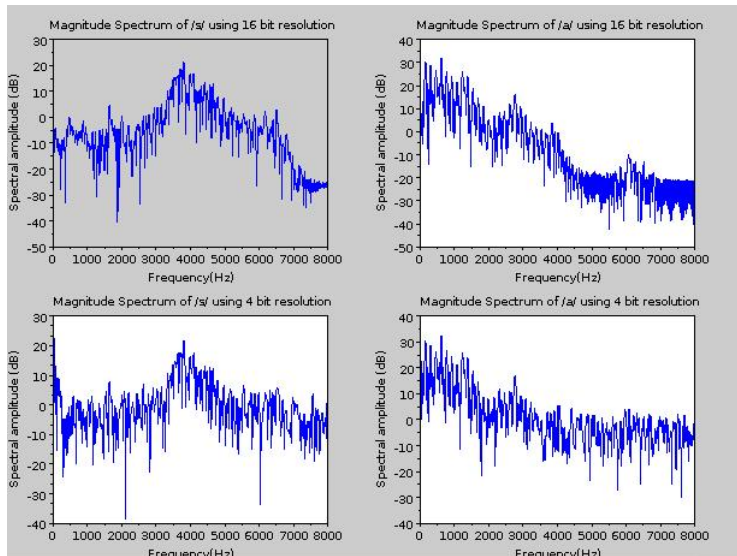
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Speech Signal at 16 vs 4 bit resolution



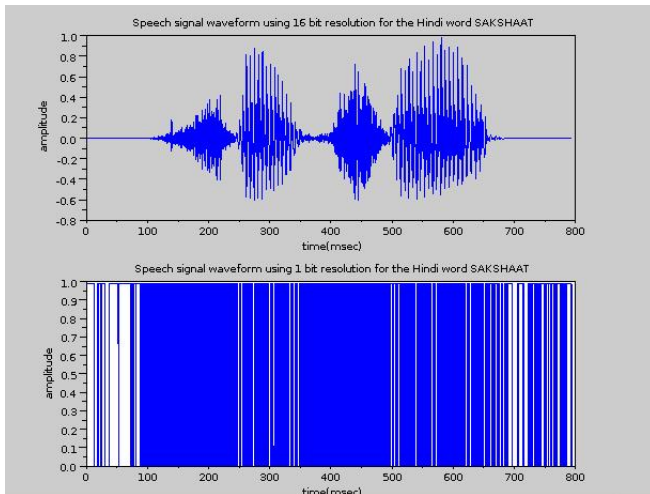
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Speech Spectra at 16 vs 4 bit resolution



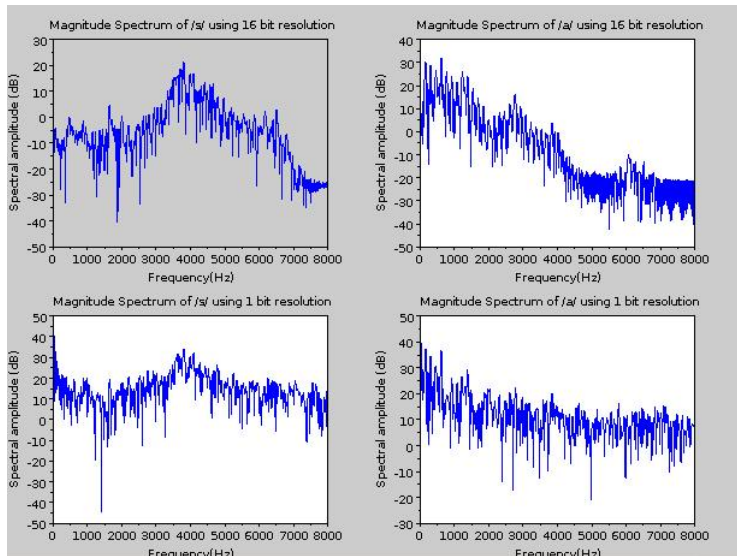
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Speech Signal at 16 vs 1 bit resolution



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Speech Spectra at 16 vs 1 bit resolution

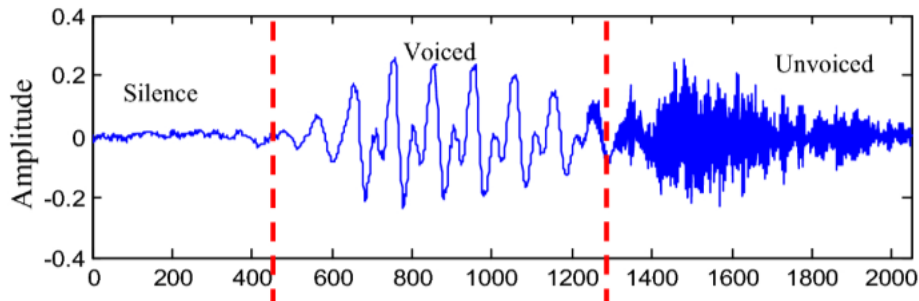


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Broad Segmentation of Speech

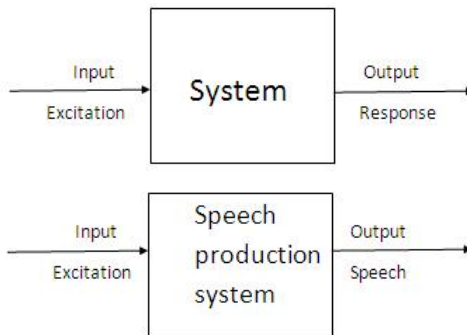
- **Three Regions** : Voiced, unvoiced and silence regions
- **Voiced** : Segments / regions of speech produced by voiced excitation (vibration of vocal folds)
- **Unvoiced** : Segments / regions of speech produced by unvoiced excitation (burst, frication, aspiration)
- **Silence** : Segments / regions of speech during which there is absence of any type of excitation

Voiced-Unvoiced-Silence Speech Segments



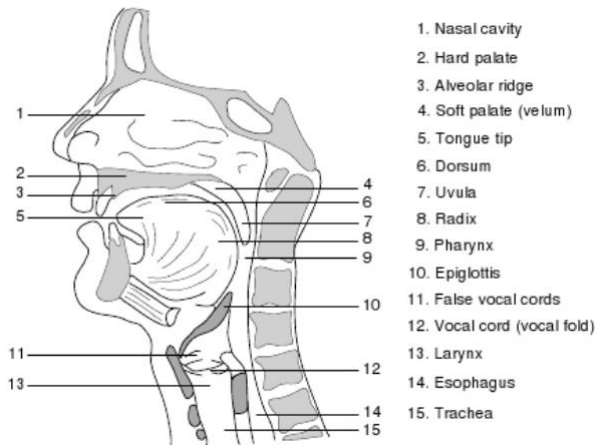
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Signal and System



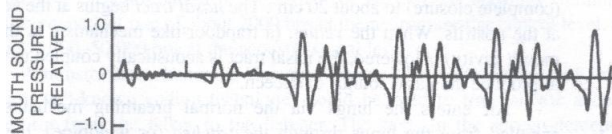
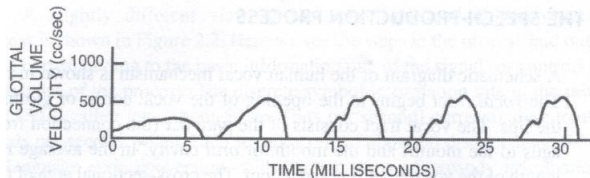
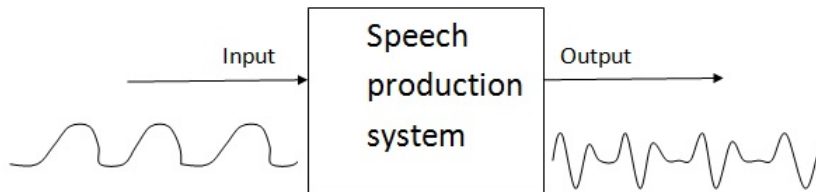
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Human Speech Production System



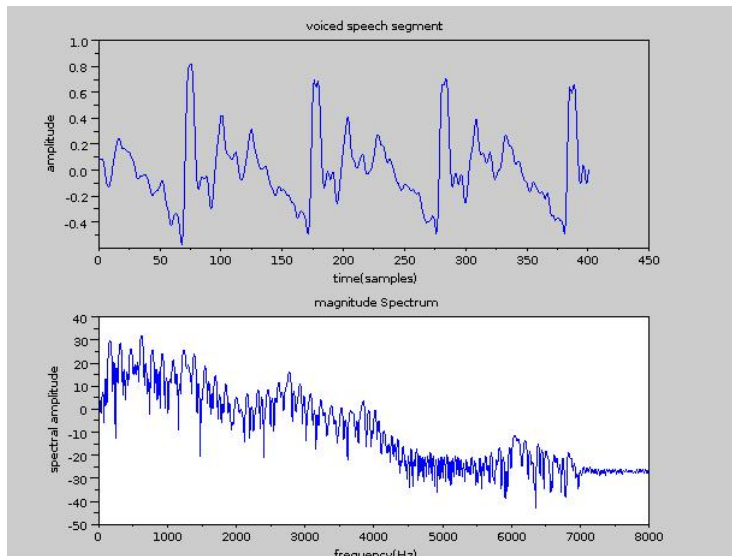
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Voiced Speech Production



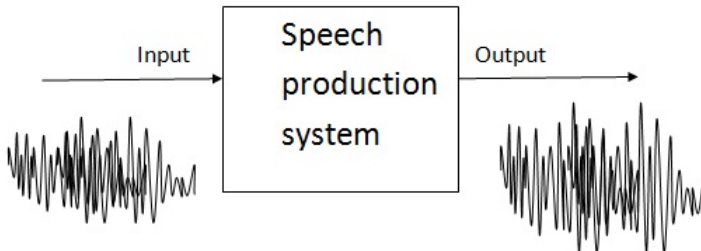
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Voiced Speech Segment



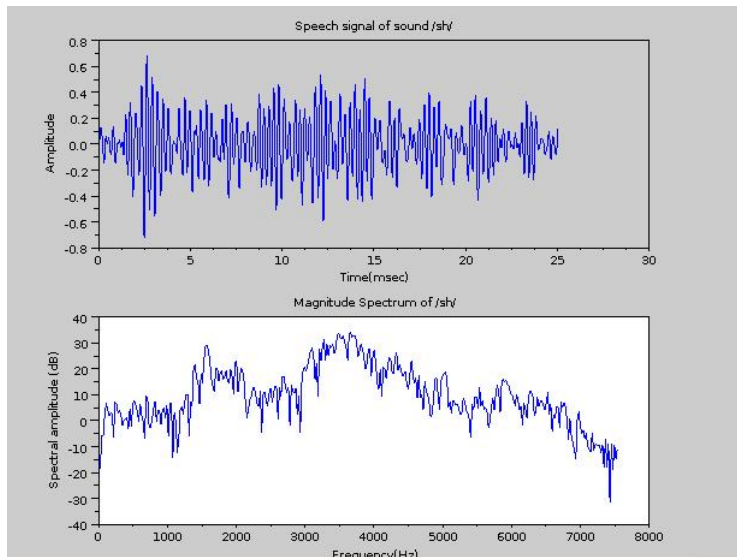
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Unvoiced Speech Production



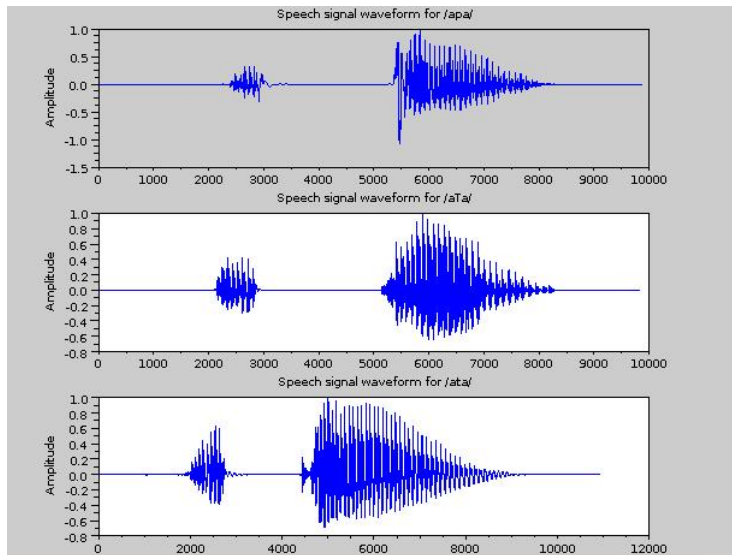
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Unvoiced Speech Segment



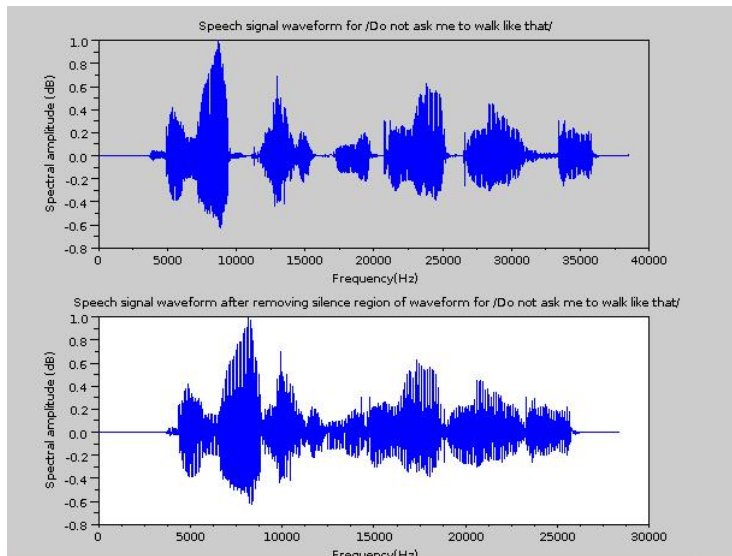
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Silence Regions in Speech



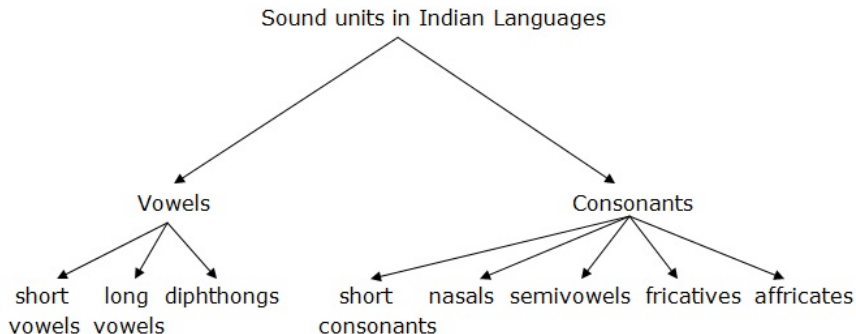
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Silence Regions in Speech



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Hindi Sound Units Classification



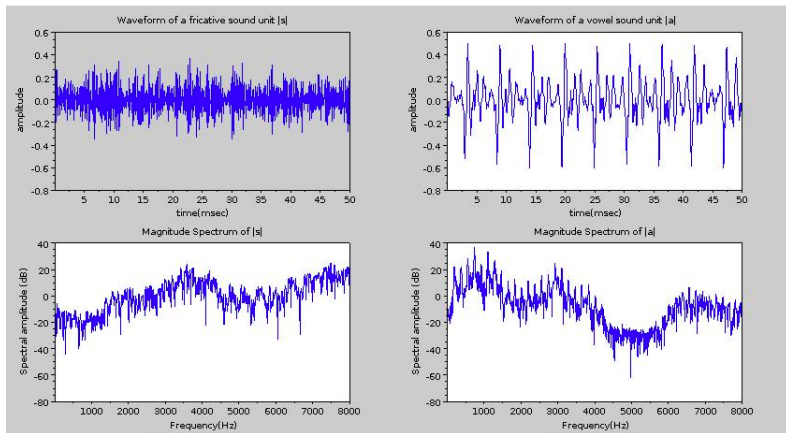
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Hindi Sound Units

Vowels		Consonants				
अ	a	क	ख	ग	घ	ङ
आ (ा)	A	k	kh	g	gh	n
इ (ि)	i	च	छ	ज	झ	ञ
ई (ि)	I	c	ch	j	jh	n
उ (उ)	u	ट	ठ	ड	ढ	ण
ऊ (ू)	U	T	Th	D	Dh	N
ऋ (ृ)	R	त	थ	द	ध	न
ॠ (ॡ)	RR	t	th	d	dh	n
लृ (ॢ)	lr.	प	फ	ब	भ	म
ए (े)	e	p	ph	b	bh	m
ऐ (ै)	ai	य	र	ल	व	
ओ (ो)	o	y	r	l	v/w	
औ (ौ)	au	श	ष	स	ह	ज्ञ
अं (ं)	m.	S	sh	s	h	jn
अः (ः)	:					

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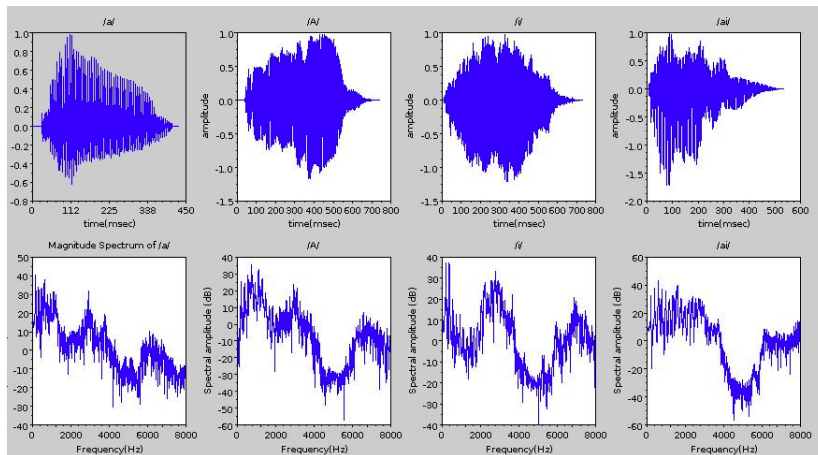
Time & Freq Domain Repn of Vowels and Consonants



- Nearly voiced vs unvoiced excitation
- Low frequency vs Low to high frequency

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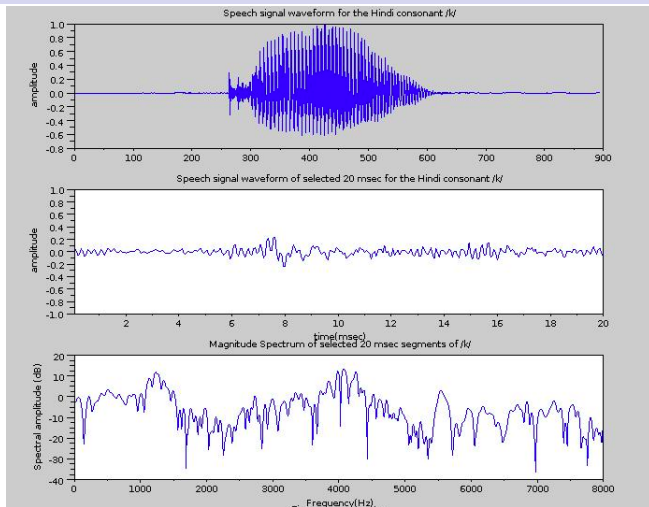
Time & Freq Domain Repn of Short & Long Vowels



- Except time duration no difference between short & long versions

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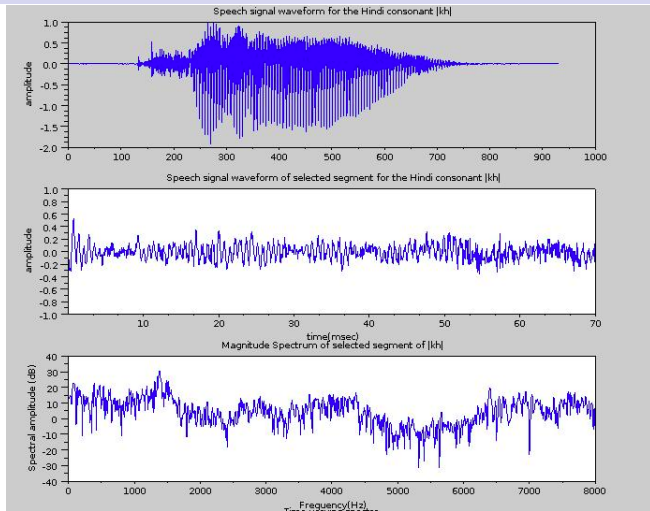
Time & Freq Domain Repn of Velar Stop Consonant



- Stop consonant is low energy region prior to vowel

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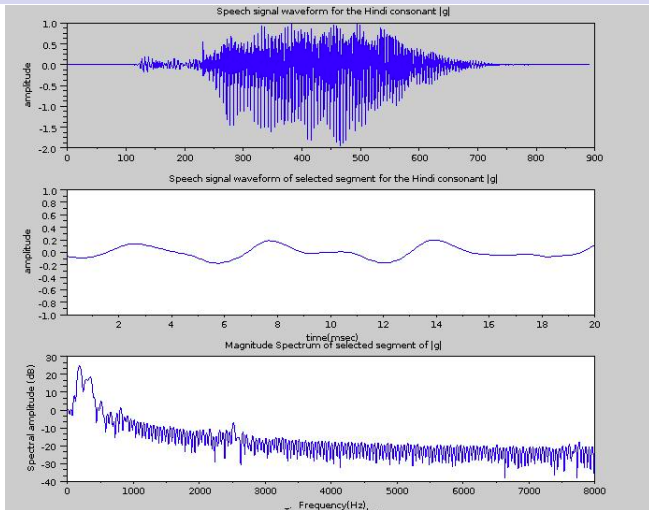
Time & Freq Domain Repn of Velar Aspirated Consonant



- Low energy noisy region prior to vowel

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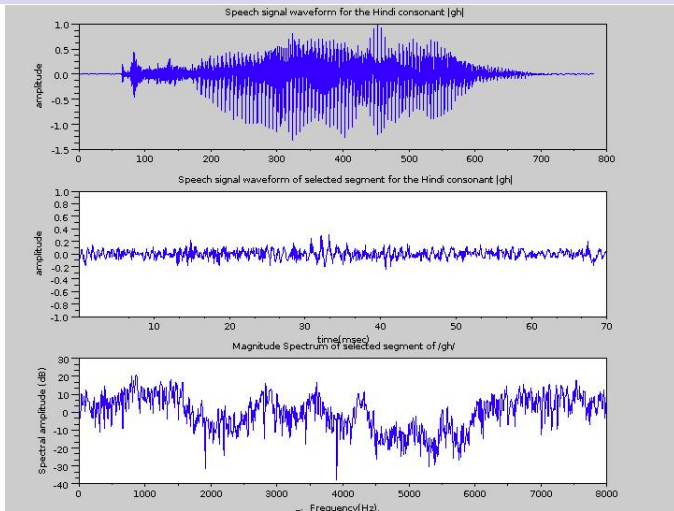
Time & Freq Domain Repn of Velar Voiced Consonant



- Low energy nearly periodic region prior to vowel

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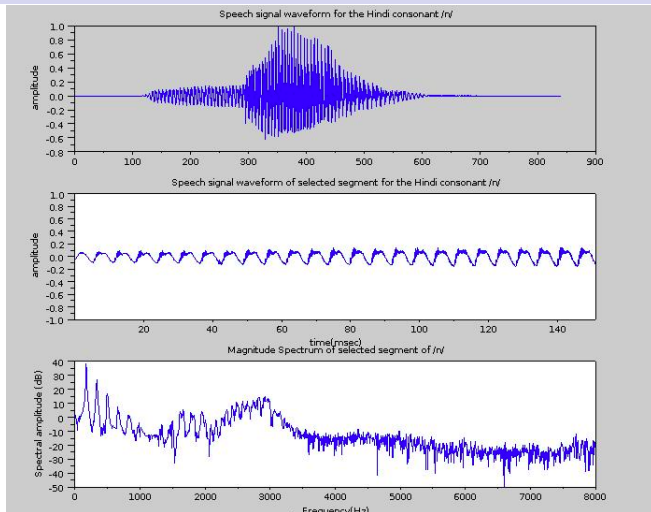
Time & Freq Domain Repn of Velar Voiced Aspirated



- Noisy region with underlying voiced excitation prior to vowel

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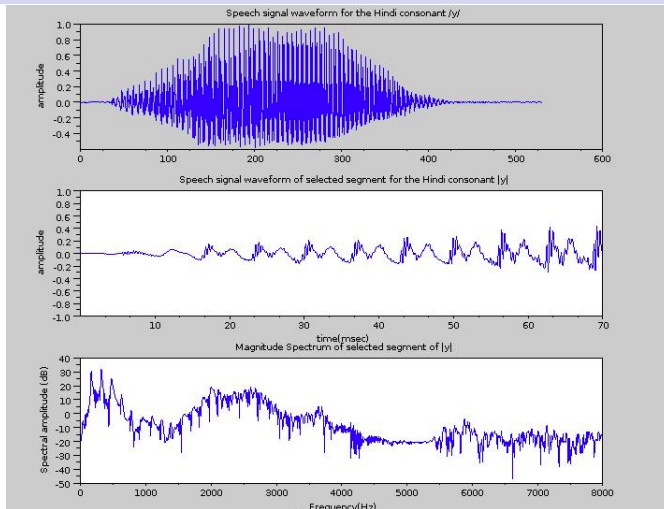
Time & Freq Domain Repn of Nasal



- Relatively low energy voiced region prior to vowel

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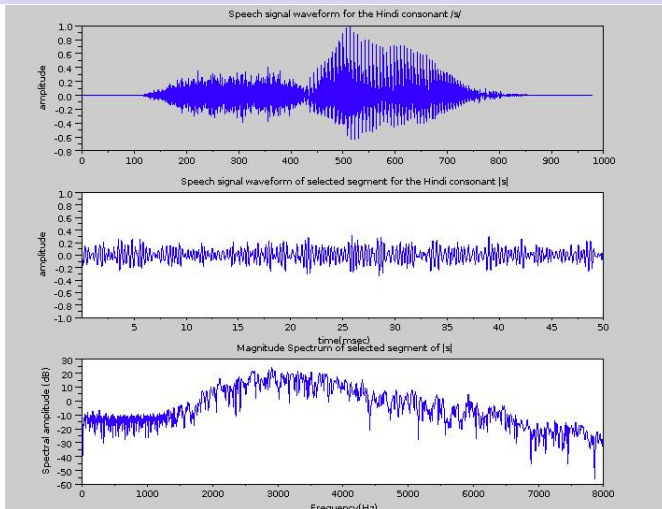
Time & Freq Domain Repn of Semivowel



- Similar to vowel in TD & FD

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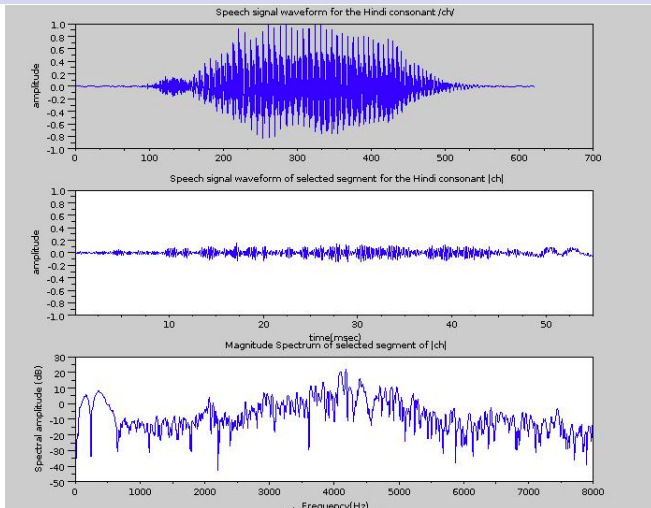
Time & Freq Domain Repn of Fricative



- Noisy structure prior to vowel

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Time & Freq Domain Repn of Affricate



- Shorter noisy structure compared to fricative prior to vowel

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Thank You